

Comparative Phylogeographic Analysis of Highly Pathogenic Avian Influenza H5 Outbreaks in South Korea Revealed Changed Dynamics and Transmission Patterns

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Introduction

- In South Korea, highly pathogenic avian influenza (HPAI) H5 outbreaks occurred in clades 2.5, 2.2, 2.3.2.1, and 2.3.2.1c from 2003 to 2011.
- Since 2014, clades 2.3.4.4b, 2.3.4.4c, and 2.3.4.4e have caused more frequent and prolonged outbreaks, with clade 2.3.4.4b persisting from 2016 to 2024, except in 2019.
- Except for clades 2.5 and 2.3.4.4e, other clades showed transcontinental spread by wild birds.
- The long-persisting clade 2.3.4.4b has successfully spread to North and South America and is causing infections not only in avian species but also in various mammals including dairy cow.
- We conducted phylogeographical and statistical analyses to compare transmission patterns across each transnationally spread clades.
- Additional statistical analysis of viral transmission among locations was focused on clade 2.3.4.4b.
- We also analyzed the genomic constellation of clade 2.3.4.4b H5N1 in South Korea and compared it with viruses isolated from the other region.

Materials & Methods

Phylogeographic analysis

- HA gene sequences of each clades were obtained from GISAID.
- We down-sampled the sequences of each clade to a range of 400-600 by selection representative sequences from cluster with 99-99.9% identity.
- The time-scaled maximum clade credibility tree was constructed in each clades.
- To estimate the geographical viral transmission, we conducted Bayesian phylodynamic analysis and identified the statistical significance of each transition using Bayes Factor and posterior probability.
- We count the number of clade 2.3.4.4b introduction to Japan and South Korea where share viral outbreaks because of the geometric closeness and calculated tMRCA of each introduction.

Genotyping Clade 2.3.4.4b H5N1

- We obtained genomic sequences of LPAI and clade 2.3.4.4b H5N1 viruses which contain all 8 gene segments from GISAID.
- In our phylogenetic analysis using all Korean sequences and down-sampled sequences from other regions with 99-99.9% identities using CD-HIT.
- We constructed ML tree of each gene segments to define genotypes of clade 2.3.4.4b H5N1.

Result & Conclusion

- In outbreaks before 2014, viral outbreaks in South Korea were sustained less than 1 year. Following the emergence of clade 2.3.4.4b, viral outbreak of the clade 2.3.4.4b have persisted for more than 7 years except 2019. However, the number of outbreak cases and culled poultry has been decreased, except 2020/2021 when there was a temporary expansion of the culling zone (**Figure 1**).
- Clade 2.2 and clade 2.3.2.1 exhibit one-way transmission patterns, whereas clade 2.3.4.4c and clade 2.3.4.4b show bidirectional transmission pattern. In clade 2.3.4.4c, the first introduction case of Gs/GD lineage HPAI H5 to North America was detected. Moreover, clade 2.3.4.4b showed largest transmission boundary including South America (**Figure 2**).
- Since 2021, there has been a tendency for viruses from the same region to cluster together in the phylogenetic tree (**Figure 3a, 3b**). This suggesting clade 2.3.4.4b is becoming endemic in the scale of continent.
- We found total 10 introductions into Japan - South Korea region. Before 2017/2018 outbreak, closet region that caused viral introduction was estimated as Asia. Since 2017/2018, both Europe and Asia contributed to viral outbreaks in Japan – South Korea region with one instance from North America in 2022 (**Figure 3c, 3d**).
- We identified a total of 63 clade 2.3.4.4b H5N1 genotypes: 16 were exclusive to Asia, 31 to Europe, 13 to North America, and 1 to Africa. This result may have been underestimated due to sampling bias effect an (**Figure 4a**). In South Korea, 15 N1 genotypes and 1 N6 genotype have been identified (**Figure 4c**).
- Clade 2.3.4.4b H5N1 were actively reassorted with European LPAI generating various genotypes. Distinct genotypes have emerged in Asia, Africa, and North America, reinforcing the idea of clade 2.3.4.4b is becoming endemic.
- Genotype EA-2020-C has persisted the longest and been globally disseminated, with most genotypes derived from it. Genotype AS-2021-A and its derivatives have different NA genes from EA-2020-C (**Figure 4a, 4c**).
- In Asian countries, during the winter of 2021-2022, AS-2021-A and AS-2021-C was predominant, whereas AS-2022-AE became dominant in the winter of 2022-2023 (**Figure 4d**). This variations in the dominant genotypes between Europe and Asia demonstrate that clade 2.3.4.4b has undergone an independent evolutionary process in Asia.
- The North American virus detected in South Korea 2022 was identified as genotype B3.2. Genotype B3.13 which caused infection in diary cow in North America was not found in South Korea.
- Unlike the early epidemic, clade 2.3.4.4b is becoming enzootic and generating different genotype in each geographical region. HPAI clade 2.3.4.4b is expected to continue evolving , making it essential to monitor intercontinental virus transmission and conduct genotype analysis. Furthermore, close surveillance of the introduction of previously unobserved genotypes from other geographical location will be critical.

Result

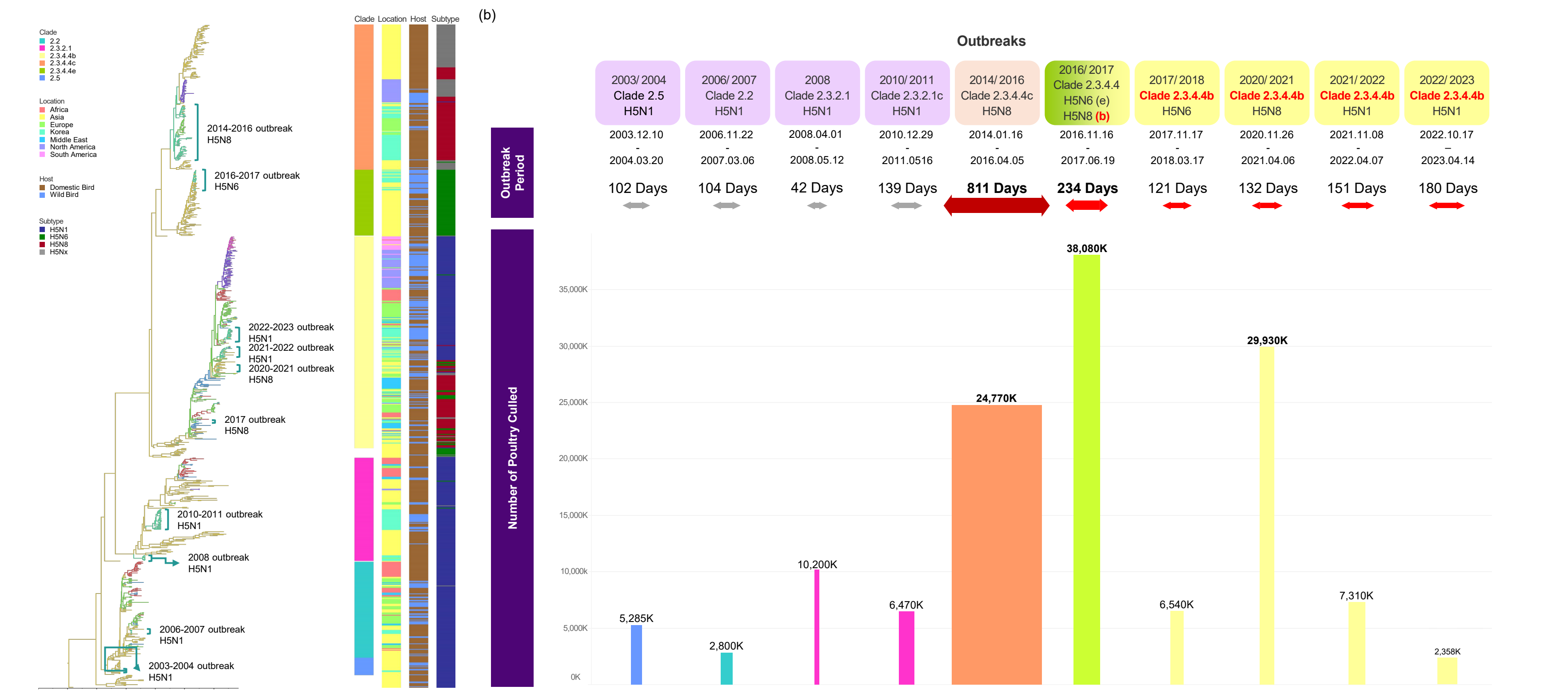


Figure 1. Outbreak history and scale of damage from the Highly Pathogenic Avian Influenza subtype H5 in South Korea. (a) The time scaled MCC tree that depicts HPAI clades in South Korea, with clades, locations, hosts, and subtypes annotated by color strips. (b) Graphical representation of outbreak periods, clade types, and the number of poultry culled during HPAI outbreaks in South Korea from 2003 to 2023

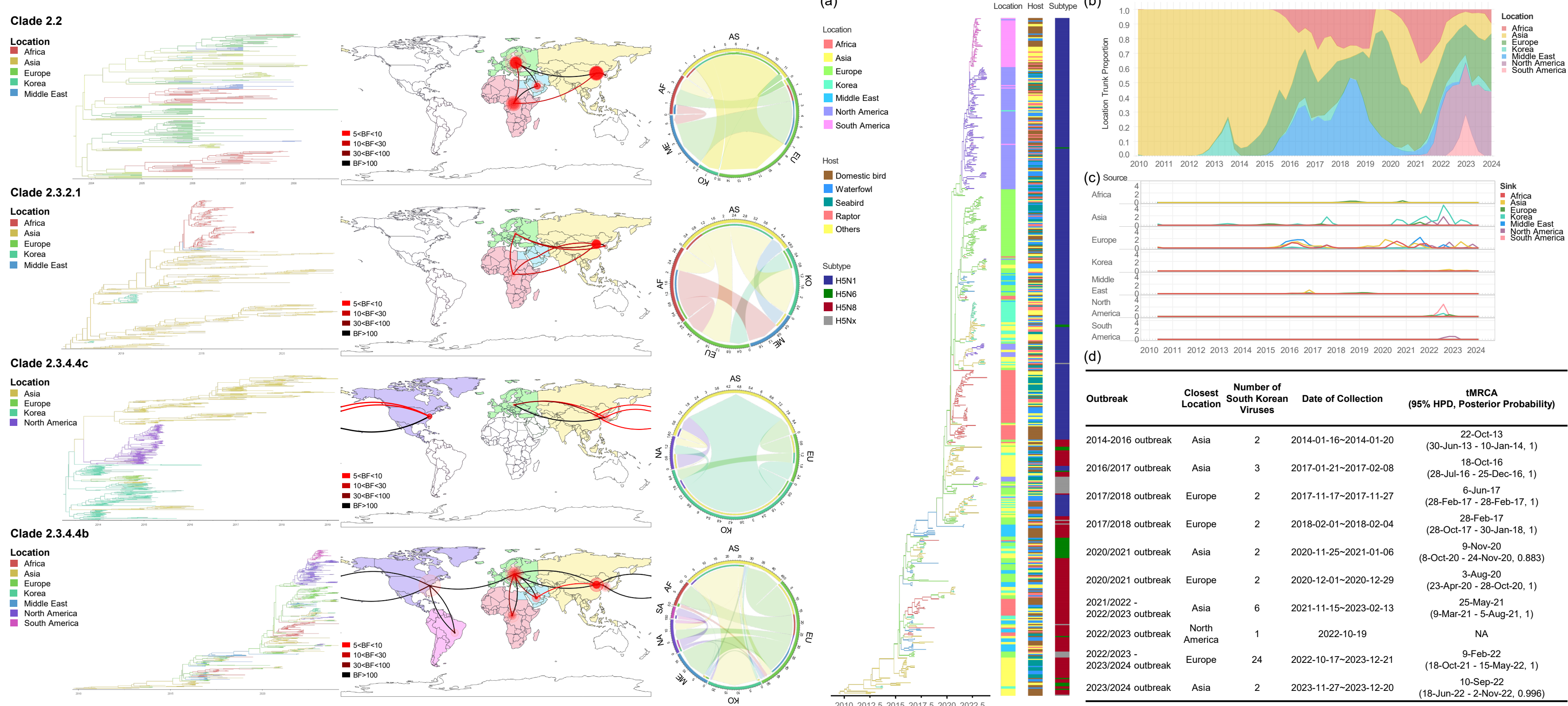


Figure 2. Time-scaled MCC tree-based Bayesian analysis of the HA gene from clades 2.2, 2.3.2.1, 2.3.4.4b, and 2.3.4.4c. Tree branches are colored by location. Phylogeographic transmissions with a posterior probability ≥ 0.5 are shown on the global map, with line colors representing Bayes factors. Markov jump counts between locations are displayed as chord diagrams.

Figure 3. The time scaled MCC tree-based Bayesian analysis of HPAI clade 2.3.4.4b HA gene. (a) Tree branch was colored by location and locations, hosts, and subtypes are annotated. (b) The trunk proportion of the location over time. (c) The chart in line format shows the number of transition events overtime. (d) tMRCA of each cluster that introduced to South Korea.

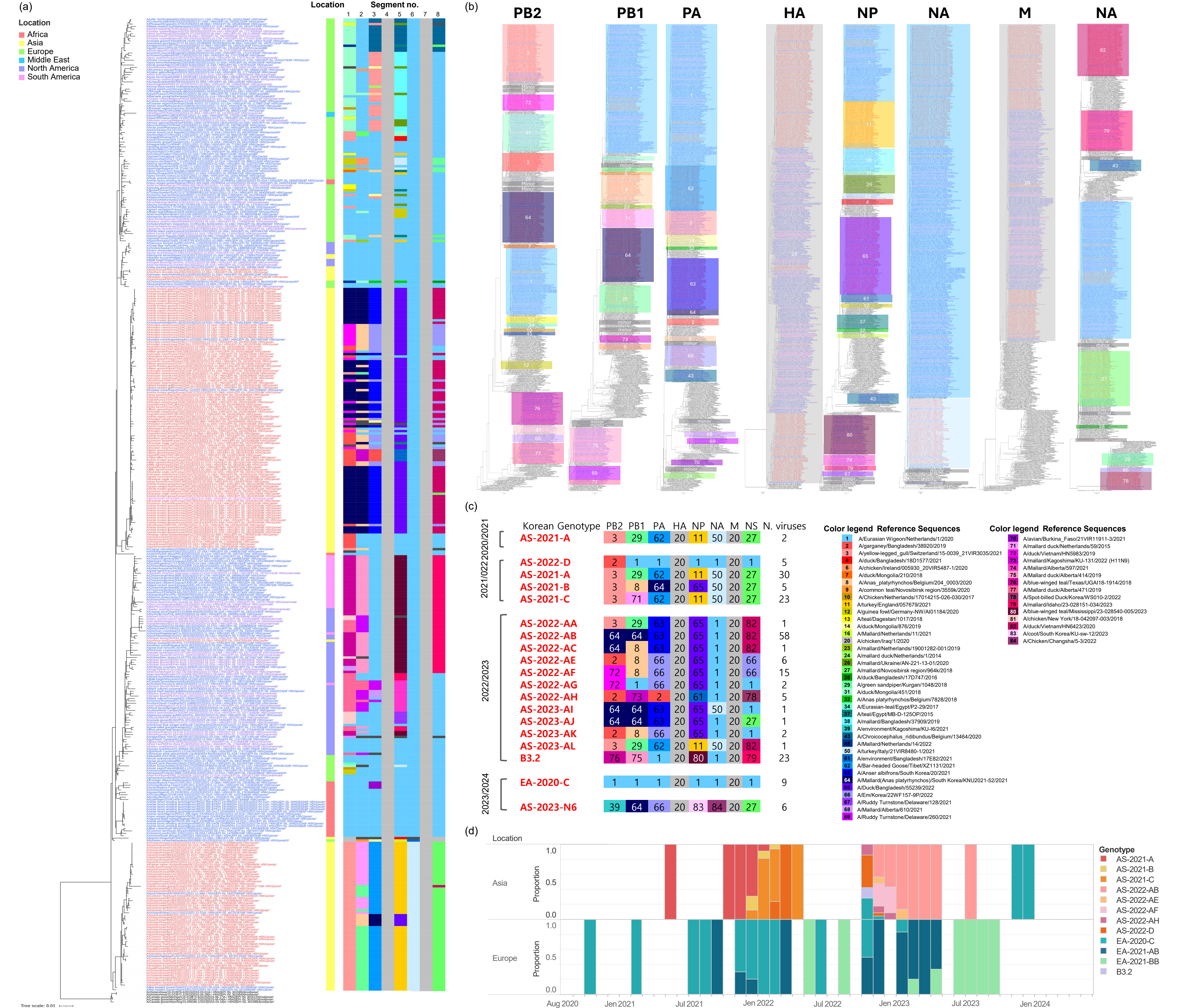


Figure 4. Graphical representation of genotypes of clade 2.3.4.4b H5N1. (a) ML tree of clade 2.3.4.4b H5N1 HA gene. South Korean avian sequences are shown in red, South Korean mammalian in pink, foreign avian in blue, and foreign mammalian in purple. Locations and references of each gene segments are annotated. (b) ML tree of each gene segments colored by reference. (c) Genotypes isolated from South Korea from 2021 to 2024 with references represented by unique numbers. (d) Monthly distribution of Asian and European genotypes containing more than 5 viruses.